

Classical Machine Learning Assignment Report

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1 Introduction

In this report we apply classical machine learning techniques to a subset of the diabetic patient records studied by Strack *et al.* [1].

The concrete task addressed in this report is a binary classification problem: given the information collected during a hospital encounter for a diabetic patient, predict the value of the `readmitted` label (YES / NO). The data provided contains 5000 patients described by 37 candidate features that mix demographics (race, gender, age range), encounter information (admission and discharge codes, length of stay, number of procedures and medications), diagnosis codes (ICD-9), and indicators of diabetes-specific medications.

2 Dataset Description

Exploring the dataset revealed that it contains 5000 rows and 38 columns. The target variable is `readmitted`, which indicates whether the patient was readmitted to the hospital in the future. The dataset rows and columns are described in Table 1.

Table 1: Variables in the raw dataset (5000 patients $\times$ 38 columns). The 15 medication columns share the same type and meaning, so they are grouped into a single row.

Column(s)	Type	Description
Unnamed: 0, patient_nbr	int	Row index and patient identifier
race, gender	cat.	Demographics
age	cat.	Age range, ordinal ([0-10) ... [90-100))
admission_type_id	int	Coded type of admission
discharge_disposition_id	int	Coded discharge disposition
admission_source_id	int	Coded source of admission
time_in_hospital	int	Length of stay (days)
num_lab_procedures	int	Number of lab procedures
num_procedures	int	Number of non-lab procedures
num_medications	int	Number of distinct medications
number_outpatient	int	Outpatient visits in the previous year
number_emergency	int	Emergency visits in the previous year
number_inpatient	int	Inpatient visits in the previous year
number_diagnoses	int	Number of diagnoses entered
diag_1, diag_2, diag_3	float	Primary and secondary diagnoses (ICD-9 codes)
diag_4	float	Extra column, not in the original dataset
15 medication columns	cat.	Drug status: No/Down/Steady/Up
change	cat.	Whether the medication was changed
diabetesMed	cat.	Whether any diabetes medication was prescribed
readmitted	cat.	Target: YES / NO

As we have said, the column to predict is `readmitted`. We check how balanced the two classes are (YES/NO), because a strong imbalance would change which metrics and techniques are

appropriate. We find 2718 NO (54%) and 2282 YES (46%), which is balanced. So we can proceed with standard classification techniques without needing special handling for class imbalance.

3 Data Exploration and Pre-processing

This dataset uses the string '?' as a placeholder so we inspect every categorical column to find these placeholders. Doing this analysis we find that the columns `race` and `tolbutamide` contains this placeholder. In the case of `race` we have 112 entries with '?', which is 2.24% of the data. Since this column already has an 'Other' category, we merge the unknown patients into it, preserving all 5000 rows. In the case of `tolbutamide` we have 3993 entries with '?', which is 79.86% of the data. As this column only contains '?' and 'No', replacing the placeholder would yield a near-constant column, so we just drop the column as it does not provide any information.

Next step is to check for other constant or almost constant columns. Here we find that the columns `acetoexamide`, `trogliatone`, `examide` and `metformin-rosiglitazone` are constant, so we can drop them. Also the columns `chlorpropamide`, `tolazamide` and `glipizide-metformin` are almost constant, so we can drop them as well. We can also drop `Unnamed: 0` and `patient_nbr` as they are just identifiers and do not provide any useful information for the prediction.

There is a suspicious column called `diag_4` that was not in the original dataset description. Comparing for example with the `diag_3` column, the diagnosis codes are typically integers, while for `diag_4` we have floats. As we can see in Table 2, `diag_4` is essentially a noisy duplicate of `diag_2` (correlation ≈ 0.895) and is uncorrelated with the target. Since it is not in the original dataset and is highly redundant with `diag_2`, we drop it.

Table 2: Pearson correlation of `diag_4` with the other diagnosis columns and with the target.

Pair	Correlation
<code>diag_4</code> – <code>diag_1</code>	0.009
<code>diag_4</code> – <code>diag_2</code>	0.895
<code>diag_4</code> – <code>diag_3</code>	0.084
<code>diag_4</code> – <code>readmitted</code>	0.014

For these diagnostic columns we have a lot of different values, so we use the standard ICD-9 chapter ranges¹ to map each code to a broad medical category, producing 16-category columns that we name as `diag_x_cat`.

After this pre-processing step we are left with 26 feature columns, and we are ready to encode the feature (x) variables. We distinguish three types of features:

- Numeric features(8): standardised with `StandardScaler`.
- Ordinal feature (1): the `age` range, encoded with `OrdinalEncoder` following the natural order of the age intervals.
- Nominal features (17): one-hot encoded with `OneHotEncoder`.

The target variable (y) is binary, so we encode it as 0 for NO and 1 for YES.

¹ICD-9 chapter ranges: https://en.wikipedia.org/wiki/List_of_ICD-9_codes.

4 Results

4.1 Experimental setup

Before training any model, we split the data into a training set (80%, 4000 patients) and a test set (20%, 1000 patients). The split is *stratified* on the target so both sets keep the same class balance, and a fixed random seed (`random_state = 42`) makes the whole experiment reproducible.

We compare a majority-class baseline against seven classical classifiers:

- Dummy classifier: always predicts the majority class (NO). It only provides a floor that any useful model must beat.
- Logistic regression: a linear model that estimates the probability of the positive class.
- k -nearest neighbours (KNN): classifies each patient by the majority vote of its closest neighbours.
- Gaussian naive Bayes: a probabilistic classifier that assumes the features are independent given the class.
- Decision tree: a single tree of yes/no rules.
- Support vector machine (SVM): finds a maximum margin boundary in a non-linear feature space.
- Random forest: an ensemble that averages many decision trees trained on bootstrap samples.
- Gradient boosting: an ensemble that builds trees sequentially, each one correcting the errors of the previous ones.

Each model is evaluated with 5 fold stratified cross validation on the training set. We report three metrics: accuracy (fraction of correct predictions, meaningful here because the classes are balanced), the F1-score of the positive class (the harmonic mean of precision and recall for the YES class), and ROC-AUC (the area under the ROC curve, a threshold independent measure of how well the model ranks patients by risk). ROC-AUC is used as the primary metric for model selection.

4.2 Model comparison

Table 3 reports the cross-validated results, and Figure 1 overlays the ROC curve of every model.

Table 3: 5-fold stratified cross-validation results on the training set (mean $\pm$ std), sorted by ROC-AUC.

Model	Accuracy	F1 (YES)	ROC-AUC
SVM (RBF)	0.632 $\pm$ 0.009	0.484 $\pm$ 0.010	0.677 $\pm$ 0.015
Gradient Boosting	0.633 $\pm$ 0.008	0.539 $\pm$ 0.013	0.674 $\pm$ 0.010
Logistic Regression	0.630 $\pm$ 0.018	0.539 $\pm$ 0.021	0.672 $\pm$ 0.011
Random Forest	0.624 $\pm$ 0.017	0.544 $\pm$ 0.016	0.665 $\pm$ 0.015
KNN	0.573 $\pm$ 0.012	0.481 $\pm$ 0.023	0.599 $\pm$ 0.010
Naive Bayes	0.493 $\pm$ 0.006	0.636 $\pm$ 0.004	0.557 $\pm$ 0.009
Decision Tree	0.551 $\pm$ 0.021	0.511 $\pm$ 0.019	0.548 $\pm$ 0.020
Dummy (majority class)	0.544 $\pm$ 0.000	0.000 $\pm$ 0.000	0.500 $\pm$ 0.000

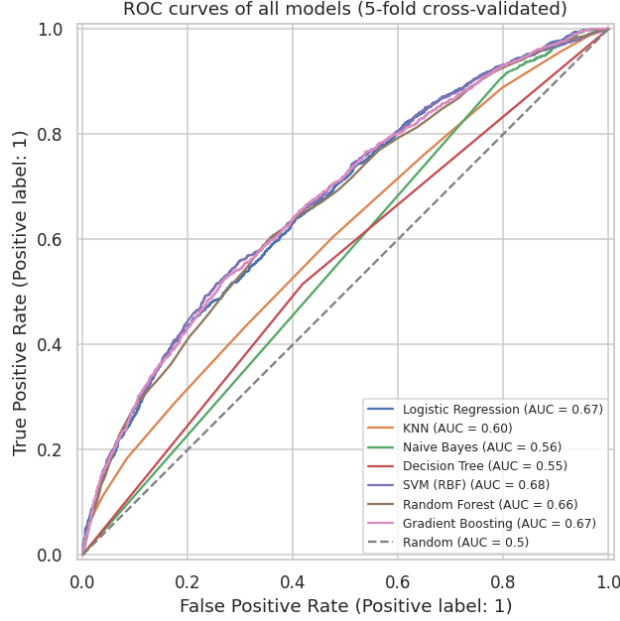


Figure 1: ROC curves of all models, computed from cross-validated predictions on the training set. The dashed diagonal corresponds to a random classifier.

Three groups of models can be distinguished. The strongest group (SVM , gradient boosting and logistic regression) reaches a ROC-AUC of about 0.67-0.68. The random forest, with its default hyper-parameters, performs slightly below them. Finally, KNN, naive Bayes and the decision tree form a clearly weaker group. The decision tree overfits the training folds and ends up close to random (ROC-AUC 0.55). Naive Bayes shows an interesting pattern: it has the lowest accuracy - (even below the majority class baseline+) yet the highest F1-score, because it predicts the positive class very often (high recall, but many false positives). Its feature independence assumption is badly violated by the one-hot encoded columns. With the exception of the decision tree, every model clearly beats the baseline ROC-AUC of 0.5, confirming that they extract real signal from the data.

4.3 Hyper-parameter tuning

The two ensemble models (random forest and gradient boosting) were the most promising, so we tuned them with an exhaustive grid search . The explored grids are listed in Table 4.

Table 4: Hyper-parameter grids explored with GridSearchCV.

Model	Hyper-parameter	Values
Random Forest	<code>n_estimators</code>	200, 400
	<code>max_depth</code>	None, 10, 20
	<code>min_samples_leaf</code>	1, 5
Gradient Boosting	<code>n_estimators</code>	200, 400
	<code>learning_rate</code>	0.05, 0.1
	<code>max_depth</code>	2, 3

The best configurations found were:

- Random forest: `n_estimators` = 400, `max_depth` = 10, `min_samples_leaf` = 1, reaching a cross-validated ROC-AUC of **0.679**.
- Gradient boosting: `n_estimators` = 200, `learning_rate` = 0.05, `max_depth` = 2, reach-

ing a cross-validated ROC-AUC of **0.676**.

Limiting the depth of the random forest (from unbounded to 10) noticeably improves it, raising its ROC-AUC from 0.665 to 0.679 and making it the best model overall. We therefore select the tuned random forest as our final model.

4.4 Final evaluation on the test set

We now evaluate the tuned random forest on the test set, which it has never seen. The results are summarised in Table 5 and Figure 2.

Table 5: Performance of the tuned random forest on the held-out test set (1000 patients).

Class	Precision	Recall	F1-score
NO (not readmitted)	0.63	0.80	0.71
YES (readmitted)	0.65	0.45	0.53
Accuracy = 0.64		ROC-AUC = 0.696	

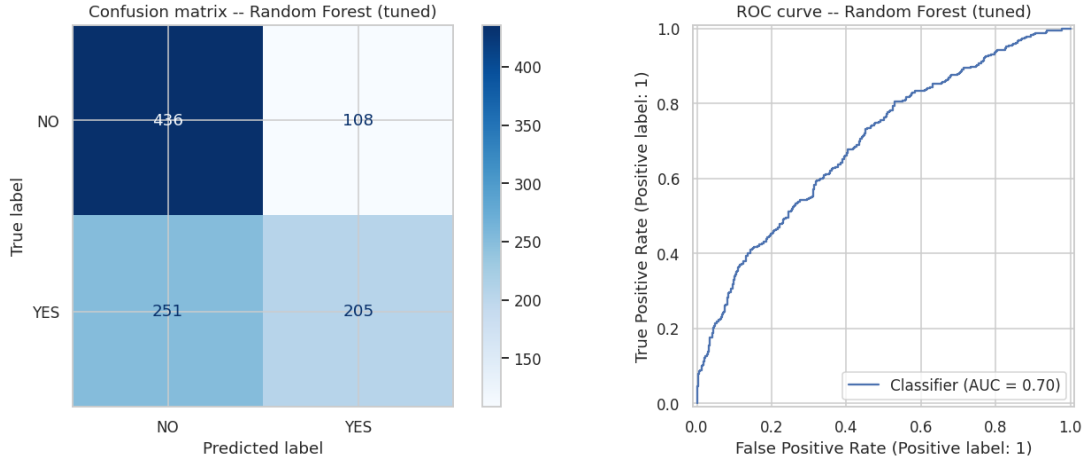


Figure 2: Tuned random forest on the test set: confusion matrix (left) and ROC curve (right).

On the test set the model reaches an accuracy of 0.64 and a ROC-AUC of 0.696, consistent with the cross-validation estimate, which means there is no sign of overfitting. The confusion matrix reveals that the model identifies non-readmitted patients well (0.80 for the NO class) but only catches about 45% of the patients who are actually readmitted. When it does predict a readmission, it is right about 65% of the time. This behaviour is typical when the available features only partially explain the outcome.

4.5 Feature importance

Finally, we inspect which features the tuned random forest relies on most, using the impurity-based importance scores. Figure 3 shows the fifteen most important features.

The single most important feature, by a large margin, is `number_inpatient` — the number of inpatient visits in the previous year. It is followed by other measures of prior hospital utilisation and encounter intensity (`num_lab_procedures`, `num_medications`, `number_emergency`, `number_diagnoses`, `number_outpatient`, `time_in_hospital`) together with `age`. This is clinically intuitive: patients who have used the hospital intensively in the recent past, who carry many diagnoses, and who are older are the most likely to be readmitted. The medication and demographic columns contribute very little.

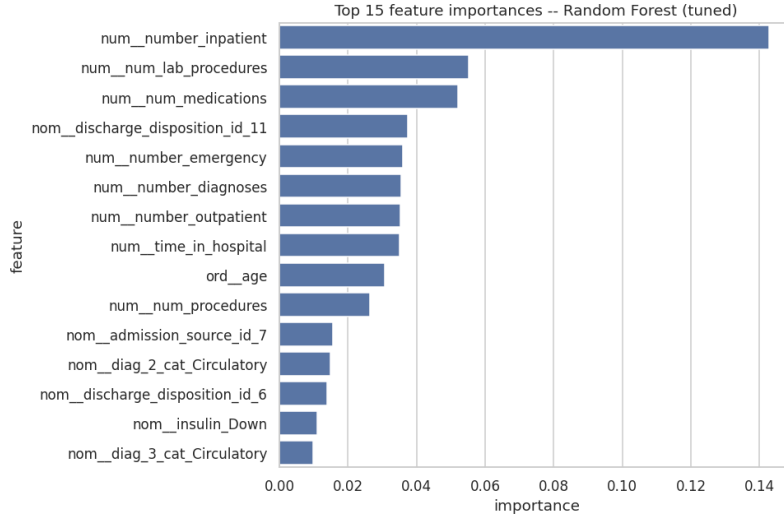


Figure 3: Fifteen most important features for the tuned random forest, measured by mean decrease in impurity. The prefixes `num_`, `ord_` and `nom_` indicate the numeric, ordinal and nominal transformer that produced each feature.

References

- [1] B. Strack, J. P. DeShazo, C. Gennings, J. L. Olmo, S. Ventura, K. J. Cios, and J. N. Clore, "Impact of HbA1c Measurement on Hospital Readmission Rates: Analysis of 70,000 Clinical Database Patient Records," *BioMed Research International*, vol. 2014, Article ID 781670, 11 pages, 2014.